

## SECTION A

### 1 (short)

- (a) State Thevenin's and Norton's theorems. [3]
- (b) Calculate the Thevenin and Norton equivalents of the circuit of Fig. 1. [7]

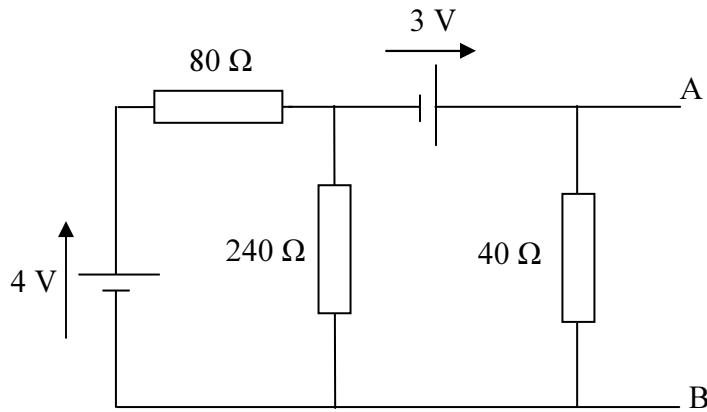


Fig. 1

### 2 (short)

- (a) State the assumptions made when an op-amp is described as being 'ideal'. [3]
- (b) The op-amp for the circuit in Fig. 2 may be assumed to be ideal. Derive an expression for the DC voltage gain of the circuit. Also find an expression for the small-signal voltage gain including the effect of  $C_2$ , and hence derive the  $-3$  dB frequency of the circuit. [7]

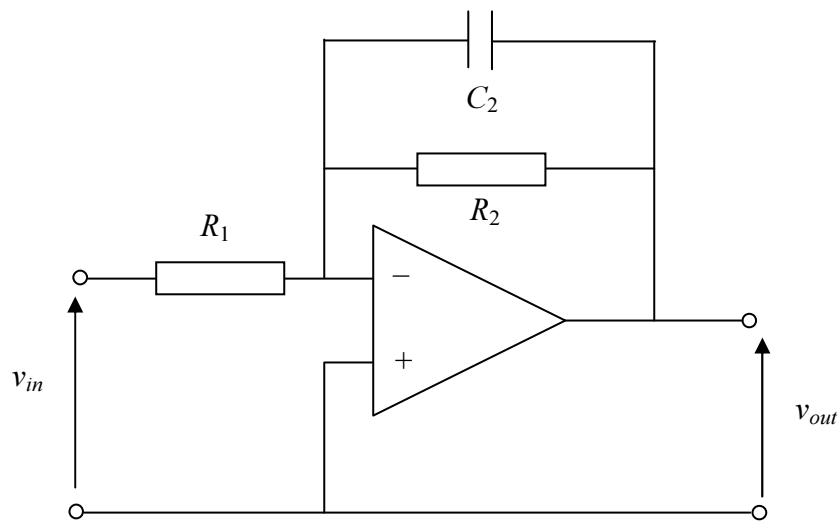


Fig. 2